

Surgery for rectal cancer

The type of operation you need will depend on the stage of the cancer and where it is in the rectum.

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Rectal cancer surgery

Surgery is the most common treatment for [rectal cancer](#). You may have other treatments before surgery to make it easier to remove the cancer. You could have 1 of the following treatments:

- [radiotherapy](#).
- [chemotherapy](#).
- chemotherapy and radiotherapy together which is called [chemoradiation](#).

Treatment before surgery also lowers the risk of the cancer coming back in the rectum or the tissues close to it.

There are things you can do [before surgery](#) to help you to improve your fitness.

[Bowel cancer](#) is a general term for cancer that starts in either the colon or rectum. It might sometimes be called colorectal cancer.

[Surgery to the bowel \(colon\)](#) is different from surgery to the back passage (rectum).

In the video below, Oncologist Rebecca and Cancer Nurse Specialist Claire explain how bowel cancer may be treated. They talk about surgery, chemotherapy, radiotherapy, immunotherapy and targeted therapies.



We understand that having treatment can be a difficult time for people. We are here to support you. If you want to talk, you can:

- call the Macmillan Support Line for free on [0808 808 00 00](tel:0808 808 00 00)
- [chat to our specialists online](#)
- visit our [bowel \(colon and rectal\) cancer forum](#) to talk to people who have been affected by bowel cancer, share your experience, and ask an expert your questions.

Before your operation

Before your operation, you go to a pre-assessment clinic to check you are fit for the operation. At the clinic, you may have [blood tests](#), a blood pressure check and a [recording of your heart \(ECG\)](#).

For some types of rectal cancer surgery, your bowel needs to be empty. A nurse may advise you about preparing for your operation. You may need to follow a special diet. You may need to take a laxative for a few days before the operation. You will also be asked not to eat or drink for a few hours before the operation.

Before the operation, you will usually be given antibiotics. These are to prevent infection. You may have them as an injection or tablets.

You will meet a member of the surgical team to discuss the operation. If you are going to have [a stoma](#), you will also meet a stoma care nurse, who will explain what is involved.

During your clinic appointment, you can ask questions or share any concerns you have about the operation. If you live alone, or care for someone else, you may need help when you go home after surgery. Tell a nurse as soon as possible. They can help to make arrangements for you. We have more information about [getting support at home after leaving hospital](#).

You will usually go into hospital on the morning of the operation. You will be given elastic stockings (TED stockings) to wear during the operation and for some time afterwards. This is to prevent [blood clots](#) in your legs.

Surgery to remove rectal cancer

There are different types of surgery for rectal cancer. The type of surgery you have will depend on:

- the [stage](#) of the cancer
- where it is in [the rectum](#)
- your general health.

Surgery to remove part or all of the rectum is called a resection.

After the operation, all the tissue that the surgeon has removed is sent to a [pathologist](#). They check it carefully to see whether there are cancer cells close to the edge. The edge is called the margin.

If the pathologist finds cancer cells in the margin, there might be a higher risk of the cancer coming back. This is not common, but if it happens you may be offered a second operation or radiotherapy.

Surgery for early stage rectal cancer

If you have a very small, [stage 1 rectal cancer](#), you may have an operation called a local excision. This is a small operation to remove the cancer and some of the healthy tissue surrounding it. The cancer can usually be removed through the anus. This means the surgeon does not have to make a cut in your tummy.

There are 2 ways your surgeon can do this.

Transanal endoscopic microsurgery (TEM)

The surgeon passes a long, flexible tube into the anus and the rectum. The tube is called an endoscope. It has a light and camera on the end. This allows the surgeon to find the tumour and remove it precisely.

Transanal minimally invasive surgery (TAMIS)

The surgeon places a small port into the anus. A port is a circular device that holds the anus open. This allows the surgeon to see the rectum clearly. The surgeon then passes a laparoscope and other instruments through the port into the rectum to remove the cancer.

Surgery for locally advanced rectal cancer

If you have [locally advanced rectal cancer](#), the type of surgery you have depends on:

- where the cancer is in the rectum
- the size of the tumour
- how far it is from [the anus](#).

Open or laparoscopic surgery

Your operation may be done as either:

- open surgery
- keyhole surgery which is also called laparoscopic surgery.

Open surgery

Open surgery means the surgeon makes 1 large cut, usually near the tummy button. The length of the cut depends on the size of the cancer and where it is.

Laparoscopic (keyhole) surgery

Laparoscopic surgery means the surgeon makes 4 or 5 small cuts in the tummy. They pass a thin tube through 1 of the cuts. The tube is called a laparoscope. It has a small camera on the end. The surgeon then passes specially designed surgical tools through the other cuts. The surgeon uses these tools to remove the cancer.

Laparoscopic surgery is also sometimes called minimally invasive surgery. You usually recover more quickly from it than from open surgery.

Robotic surgery

Robotic surgery is when laparoscopic surgery is helped by a machine. The laparoscope and the surgical equipment are attached to robotic arms. The surgeon controls the robotic arms, which can move very precisely. This is very specialised surgery and only available in some hospitals in the UK.

Your surgeon will talk to you about which type of surgery is likely to be best in your situation.

Stomas (colostomy or ileostomy)

Sometimes during surgery to remove the cancer, the surgeon needs to create [a stoma](#). This is an opening the surgeon makes through the tummy (abdominal) wall. It connects the bowel to the surface of the tummy.

Having a stoma means stools (poo) and wind will not pass out of the rectum and anus in the usual way. Instead, they will pass out of the stoma, into a disposable bag you wear over the stoma.

If the stoma is made from an opening in the colon, it is called a colostomy. If it is made from an opening in the small bowel (ileum), it is called an ileostomy.

Stomas may be temporary or permanent.

The surgeon may make a temporary stoma to allow the bowel to heal after surgery. If you have a temporary stoma, you will usually have an operation to close the stoma and rejoin the bowel after you have finished your treatment. This operation is called a stoma reversal.

If the cancer is very low in the rectum and close to the anus, you are more likely to need a permanent stoma.

Your surgeon will tell you whether you are likely to have a stoma, and whether it will be temporary or permanent. If you need to have a stoma, you will be referred to a nurse who specialises in stoma care. They can talk to you about what to expect.

We have more information on [stomas](#).

Anterior resection

If you have cancer in the upper or middle part of the rectum, you may have an operation called an anterior resection or low anterior resection (LAR). The surgeon removes the part of bowel that contains the cancer. They then rejoin the 2 open ends of bowel. The diagrams show the part of the bowel that is removed, and how the 2 ends are rejoined.

You may have a temporary stoma after this operation. This is usually an ileostomy. You can usually have a stoma reversal a few months later.

Anterior resection

Image: The diagram shows the large bowel in the abdomen. The colon is shown in orange. The lower end of the colon and the rectum, removed during an anterior resection, are shown in white.

Anterior resection rejoined

Image: The diagram shows how the large bowel looks after an anterior resection. The colon is shown in orange. It shows how the end of the colon has been joined to the remaining rectum and anus, to replace the rectum that has been removed.

Abdominal perineal resection (APR)

If the cancer is very close to the anus, you may need an operation called an abdominal perineal resection. This is when the surgeon needs to remove the rectum and anus, to remove all the cancer. You will have a permanent stoma after this operation. This is usually a colostomy.

As well as the wound on your tummy, you will have a wound on your bottom where the anus has been closed. The anus may be closed using muscle, fat and skin from another part of the body. This is called a flap. This operation can be done as either keyhole (laparoscopic) surgery or open surgery. It depends on the size of the tumour.

Abdominal perineal resection

Image: The diagram shows the large bowel in the abdomen. The colon is shown in orange. The lower end of the colon, the rectum and the anus, are shown in white. This is the area removed during an abdominal perineal resection.

Total mesorectal excision (TME)

As well as removing the cancer using one of the operations mentioned, many surgeons also remove some of the fatty tissue around the rectum. This contains [lymph nodes](#) and blood vessels. The fatty tissue is called the mesorectum. The operation to remove it is called a total mesorectal excision (TME).

Surgery for advanced rectal cancer

There are different types of surgery for advanced rectal cancer. This will depend on where the cancer has spread to. Your cancer doctor will explain whether surgery is suitable for you.

Some people might need surgery if the cancer is causing a [blockage in the bowel](#).

If rectal cancer has spread to the [liver](#) or [lungs](#), surgery might be suitable for some people. Sometimes a liver resection to remove secondary liver tumours may be done at the same time as surgery to remove the cancer in the rectum.

If the cancer has grown into other organs nearby some people might need to have some of the organs in their pelvis removed. This operation is called a [pelvic exenteration](#). It is not suitable if the cancer has spread to organs in other parts of the body.

We have more information about [treating advanced bowel cancer](#).

After your operation

Your nurse will explain what to expect after your operation.

When you are ready they will help you to get out of bed or to sit up for a short time. You are likely to have a drip into your arm to give you fluids until you are eating and drinking again. You may have a tube in your tummy, close to the wound to drain fluid into a small drainage bag or bottle.

Your recovery time will depend on whether you had open or keyhole surgery.

We have more information on [recovering after bowel cancer surgery](#).

Booklets and resources



Booklets

01 May 2024

Understanding rectal cancer

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our rectal cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

National Institute for Health and Care Excellent (NICE). Colorectal cancer. NICE guideline [NG151]. Updated December 2021. Available from: www.nice.org.uk/guidance/NG151 [accessed Jan 2023].

Cervantes A, Adam R, Rosello S, et al. Metastatic colorectal cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. *Annals of Oncology*. 2023 ;34(1):10-32. Available from: www.annalsofoncology.org [accessed Jan 2023].

Sanoff HK. Improving treatment approaches for rectal cancer. *New England Journal of Medicine*. 2022;386(25); 2425–2426. Available from: [www.doi.org/10.1056/NEJMe2204282](https://doi.org/10.1056/NEJMe2204282) [accessed Jan 2023].

[Cancer Information Development team](#)

Author

[Dr Paul Ross](#)

Reviewer

Consultant Medical Oncologist
[Guy's and St Thomas' NHS Foundation Trust](#)

Date reviewed

Reviewed: 01 May 2024 | Next review: 01 May 2027

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

 Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Bowel (colon and rectal) cancer forum

An anonymous network of people affected by colon, rectal, small bowel, or other colorectal cancers, which is free to join. Share experiences, ask questions and talk to people who understand.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help